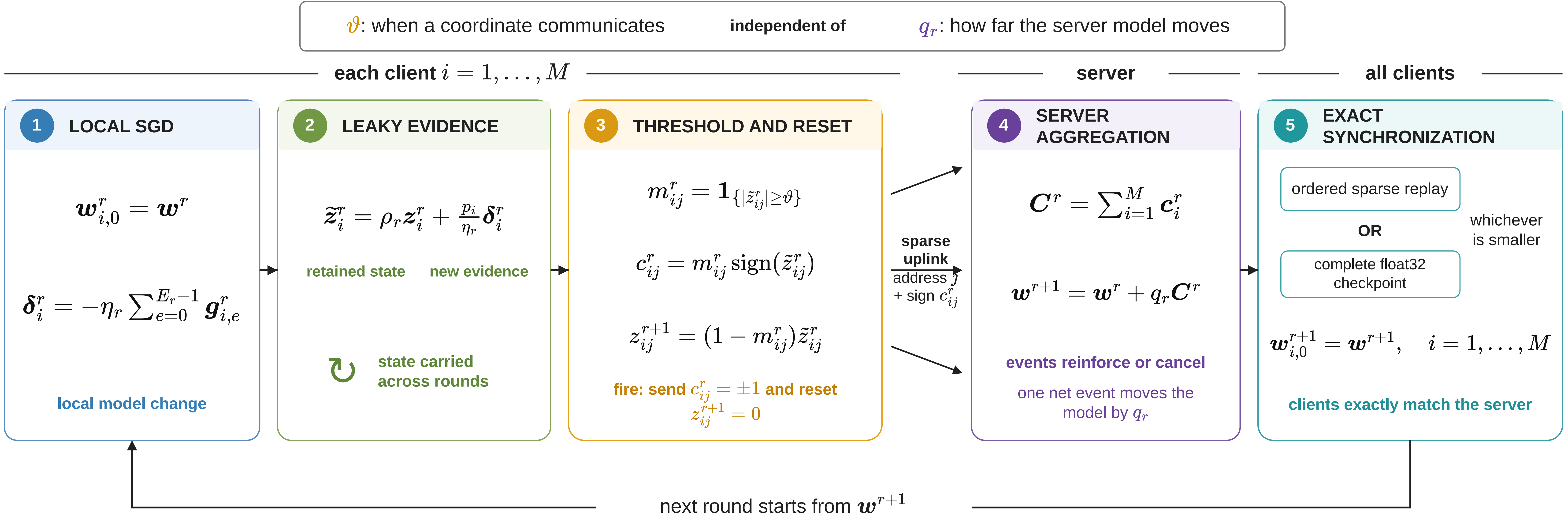

ϑ: when a coordinate communicates
independent of
q_r: how far the server model moves
each client $i = 1, \ldots, M$
server
all clients
1 LOCAL SGD
$\boldsymbol{w}_{i,0}^r = \boldsymbol{w}^r$
$\boldsymbol{\delta}_i^r = -\eta_r \sum_{e=0}^{E_r-1} \boldsymbol{g}_{i,e}^r$
local model change
2 LEAKY EVIDENCE
$\tilde{\boldsymbol{z}}_i^r = \rho_r \boldsymbol{z}_i^r + \frac{p_i}{\eta_r} \boldsymbol{\delta}_i^r$
retained state
new evidence
state carried across rounds
3 THRESHOLD AND RESET
$m_{ij}^r = \mathbf{1}_{\{|\tilde{z}_{ij}^r| \geq \vartheta\}}$
$c_{ij}^r = m_{ij}^r \, \mathrm{sign}(\tilde{z}_{ij}^r)$
$z_{ij}^{r+1} = (1 - m_{ij}^r) \tilde{z}_{ij}^r$
fire: send $c_{ij}^r = \pm 1$ and reset $z_{ij}^{r+1} = 0$
sparse uplink
address j + sign c_{ij}^r
4 SERVER AGGREGATION
$\boldsymbol{C}^r = \sum_{i=1}^M \boldsymbol{c}_i^r$
$\boldsymbol{w}^{r+1} = \boldsymbol{w}^r + q_r \boldsymbol{C}^r$
events reinforce or cancel
one net event moves the model by q_r
5 EXACT SYNCHRONIZATION
ordered sparse replay
OR
complete float32 checkpoint
whichever is smaller
$\boldsymbol{w}_{i,0}^{r+1} = \boldsymbol{w}^{r+1}, \quad i = 1, \ldots, M$
clients exactly match the server
next round starts from $\boldsymbol{w}^{r+1}$